

CENTRAL BOARD OF SECONDARY EDUCATION — PRACTICE EXAMINATION**CLASS IX • SOCIAL SCIENCE (SUBJECT CODE: 087) • ACADEMIC SESSION 2026-27****SET C — NCERT EXEMPLAR, IN-TEXT SOURCES & TRICKY QUESTIONS****CHAPTER 02: SHAPING OF THE EARTH'S SURFACE**

TIME ALLOWED: 45 MINUTES

MAXIMUM MARKS: 25 MARKS

SERIES: CBS-IX-SST-SET-C-02

Student's Name: _____

Roll No.: _____

Class & Section: Class IX - _____

Date of Exam: _____

Invigilator's Sign: _____

General Instructions (SET C - NCERT Exemplar & Tricky Questions):

1. The question paper comprises **25 NCERT Exemplar & Critical Application Questions** of **1 mark each**.
2. Tests **in-text source boxes, fine glossary definitions, historical source extracts, and tricky boundary cases**.
3. All questions are compulsory. There is no negative marking for incorrect answers.
4. Darken the corresponding bubble on the **OMR Sheet** completely using a blue/black ballpoint pen.

**NCERT IN-TEXT & EXEMPLAR TIER:** Questions in this set explore subtle details, specialized definitions, and non-intuitive concepts drawn directly from official NCERT text margins, source boxes, and glossary entries.**Q.1 Why are composite volcanoes (stratovolcanoes) like Mt. Fuji and Mt. Vesuvius characteristically steep and explosive, whereas shield volcanoes like Mauna Loa in Hawaii have gentle slopes and effusive eruptions?**

- (A) Composite volcanoes erupt highly viscous, silica-rich andesitic/dacitic magma that traps volatile gases explosively; shield volcanoes erupt low-viscosity, fluid basaltic magma that flows freely
- (B) Composite volcanoes are filled with frozen ice, while shield volcanoes are filled with boiling acid
- (C) Shield volcanoes are constructed artificially by human engineers
- (D) Composite volcanoes erupt only during night hours

Q.2 In a meandering river, why does active erosion occur on the 'cut-bank' (outer bend) while active deposition occurs on the 'point-bar' (inner bend)?

- (A) Centrifugal inertia accelerates water velocity and hydraulic shear stress along the outer bank, while friction slows water on the inside bend, causing sediment drop
- (B) The outer bank is made of soft sugar, while the inner bank is made of iron metal
- (C) Water flows uphill on the inner bend and downhill on the outer bend
- (D) Fish swim exclusively along the inner bend to push sand

Q.3 What is the critical mechanical difference between a 'continental rift valley' (like the East African Rift) and a 'river gorge' (like the Indus Gorge at Nanga Parbat)?

- (A) A rift valley is an endogenic tectonic graben formed by crustal tensional faulting where a central block drops between parallel normal faults; a gorge is carved purely by exogenic fluvial downcutting
- (B) A rift valley is formed by falling meteorites, while a gorge is formed by wind gusts
- (C) A gorge only forms in frozen glaciers, while a rift valley forms on ocean beaches
- (D) There is no difference; both are formed by underground earthworms

Q.4 How does chemical weathering via 'hydrolysis' transform hard crystalline feldspar mineral in granite into soft clay minerals?

- (A) Hydrogen ions (H^+) in acidic water react with potassium feldspar, releasing potassium and silica ions and leaving behind hydrated aluminosilicate clay (Kaolinite)
- (B) Water boils the granite rock until it turns into molten steam
- (C) Feldspar evaporates directly into the atmosphere as a gas
- (D) Granite turns into pure metallic iron when touched by rain

Q.5 Why do earthquakes of identical Richter magnitude often produce vastly different death tolls and damage depending on local soil conditions?

- (A) Loose, water-saturated uncompacted alluvial soils undergo 'soil liquefaction' and seismic wave amplification, whereas solid bedrock transmits waves with minimal ground displacement
- (B) Earthquake waves only travel toward wooden houses and avoid brick buildings
- (C) Richter scales are deliberately calibrated to give false readings in cities
- (D) Earthquakes only affect areas where people are awake

Q.6 In Karst limestone topography, what is the chemical distinction between a 'stalactite' and a 'stalagmite'?

- (A) Both are calcium carbonate precipitates, but stalactites grow downward from the cave ceiling as water drips, while stalagmites build upward from the cave floor where drops land
- (B) Stalactites are living plants, while stalagmites are frozen icicles
- (C) Stalactites are formed by wind, while stalagmites are formed by lava
- (D) Stalactites are made of pure gold, while stalagmites are made of salt

Q.7 Why do tectonic plates continue to move across the Earth's surface despite massive frictional resistance between rock slabs?

- (A) Mantle convection currents driven by radioactive decay heat in the core, combined with 'slab-pull' (the gravitational sinking of cold, dense oceanic lithosphere at subduction zones)
- (B) Ocean waves pushing against continental shorelines like sails
- (C) Earth's magnetic field pulling iron deposits toward the North Pole
- (D) The gravitational pull of passing artificial satellites

Q.8 What is the geomorphic difference between 'weathering' and 'mass wasting' (mass movement)?

- (A) Weathering is the in-situ breakdown of rock without substantial movement; mass wasting is the downslope bulk movement of rock and soil debris under the direct influence of gravity
- (B) Weathering happens only in winter; mass wasting happens only in summer
- (C) Mass wasting is caused by commercial bulldozers; weathering is caused by birds
- (D) There is no difference; both terms describe the formation of sand dunes

Q.9 Why do 'hanging valleys' form in glaciated mountain regions, producing dramatic waterfalls like Bridalveil Fall in Yosemite?

- (A) A massive main trunk glacier cuts its valley floor far deeper than smaller tributary glaciers; when the ice melts, tributary valley mouths are left perched high above the main floor
- (B) Earthquakes lift the tributary valleys into the sky after glaciers melt
- (C) Rivers flow backward up the mountainside to create hanging pools
- (D) Hanging valleys are formed by ancient human builders using stone ramps

Q.10 In aeolian desert processes, what is the aerodynamic difference between 'saltation' and 'suspension'?

- (A) Saltation is the bouncing/hopping movement of medium sand grains (0.1–0.5 mm) along the ground; suspension is the transport of fine silt and clay particles (<0.05 mm) aloft in the air
- (B) Saltation occurs only in oceans; suspension occurs only in clouds
- (C) Saltation transports boulders weighing ten tonnes; suspension transports liquid water
- (D) Both terms refer to the sliding of massive sand dunes across rivers

Q.11 Why is the Mariana Trench in the western Pacific Ocean the deepest location on Earth (~11,000 meters)?

- (A) It is a subduction trench where exceptionally old, cold, and dense oceanic lithosphere of the Pacific Plate bends sharply and sinks steeply into the mantle
- (B) It was carved by a giant subterranean river flowing beneath the ocean floor
- (C) A giant meteorite scooped out the rock down to the earth's core
- (D) It is an open volcanic crater that erupted all its rock into space

Q.12 What is the structural difference between 'anticlinal ridges' and 'synclinal valleys' in young fold mountain chains?

- (A) Anticlines are upwarped convex arches that initially form mountain crests; synclines are downwarped concave troughs that form valley depressions
- (B) Anticlines are filled with water; synclines are filled with lava
- (C) Synclines are formed by wind; anticlines are formed by ocean waves
- (D) Both terms describe flat, undeformed horizontal plains

Q.13 Why does physical (mechanical) weathering dominate in extremely cold Arctic tundras and hyper-arid deserts, while chemical weathering is subdued?

- (A) Both environments lack abundant liquid water and warm temperatures necessary to drive rapid biochemical decomposition reactions
- (B) Arctic rocks are made of diamond; desert rocks are made of plastic
- (C) The sun never shines in deserts or in the Arctic
- (D) Chemical weathering is legally prohibited in polar nature reserves

Q.14 What is an 'alluvial fan' versus a 'delta', and where do they typically form?

- (A) An alluvial fan forms on land at the base of a mountain where stream velocity abruptly drops; a delta forms where a river enters a standing body of water (sea or lake)
- (B) An alluvial fan forms inside an ocean; a delta forms on a glacier peak
- (C) An alluvial fan is carved by wind; a delta is formed by volcanic magma
- (D) Both landforms are identical in shape, location, and hydraulic origin

Q.15 How do 'sea stacks' along coastlines eventually get reduced to 'sea stumps'?

- (A) Continuous basal wave undercutting and hydraulic pounding eventually collapse the tall rock pillar, leaving a low rock platform exposed only at low tide
- (B) Sea stacks are dissolved overnight by rainwater
- (C) Sea stacks grow wings and fly away into the ocean
- (D) Marine whales chew the stone pillar down to the water surface

Q.16 What is 'biological weathering' through the action of lichens growing on bare rock faces?

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|---|--|
| (A) Lichens secrete organic acids (oxalic and lichenic acids) that chelate mineral ions, slowly breaking down rock surfaces to form initial primitive soil | (B) Lichens drill mechanical holes through granite using wooden teeth |
| (C) Lichens heat the rock to over 1,000 °C through volcanic energy | (D) Lichens turn rocks into liquid water that evaporates |

Q.17 Why do rivers develop an 'interlocking spur' landscape in their youthful upper stage?

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| (A) The river lacks the kinetic stream power to cut through hard rock ridges laterally, so it meanders around projecting spurs of harder rock as it cuts vertically downward | (B) The river flows in a straight pipe laid by human engineers |
| (C) Earthquakes push the river channel left and right every hour | (D) The river channel is made of interlocking wooden blocks |

Q.18 What is a 'cirque' (or corrie) in alpine glacial geomorphology?

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|--|---|
| (A) A steep-walled, armchair-shaped hollow excavated high on a mountain flank by rotational glacial plucking and freeze-thaw nivation | (B) A circular sand dune formed by desert storms |
| (C) A deep underwater cave filled with coral reefs | (D) A tall volcanic chimney that erupts ash clouds |

Q.19 When water collects in an abandoned glacial cirque hollow after the ice melts, the resulting mountain tarn lake is called a:

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|----------------------|-----------------------|
| (A) Tarn lake | (B) Oxbow lake |
| (C) Lagoon | (D) Caldera |

Q.20 What is 'deflation' in desert wind processes, and how does it form a 'desert pavement' (reg)?

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| (A) Wind removes fine silt and sand particles aloft, leaving behind a residual armor of closely packed, interlocking coarse pebbles and gravels | (B) Deflation is when desert sand is pumped with pressurized air |
| (C) Deflation is a financial drop in the price of desert dates | (D) Desert pavement is constructed by civil road contractors using concrete |

Q.21 Why are 'horst' and 'graben' structures associated with tensional tectonic environments rather than compressional environments?

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|--|---|
| (A) Tensional crustal pulling creates paired normal faults where blocks subside (graben/rift valleys) while intervening uplifted blocks stand as block mountains (horsts) | (B) Tensional forces compress rock into giant circular balls |
| (C) Horsts and grabens are formed by falling hail storms | (D) Tensional forces only occur beneath frozen glaciers |

Q.22 What is 'exfoliation' (onion-skin weathering) in massive granitic dome landscapes?

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|---|--|
| (A) Removal of overlying rock releases confining pressure (unloading), causing curved outer rock sheets to expand, fracture, and peel away like onion layers | (B) Acid rain dissolves the entire mountain in twenty minutes |
| (C) Underground magma boils the surface of granite domes | (D) Vegetation pulls the mountain apart with giant vine nets |

Q.23 Why does the 'Richter scale' of earthquake magnitude differ from the 'Mercalli scale'?

- (A) Richter measures absolute instrumental seismic energy released at the focus logarithmically; Mercalli measures observed surface destruction and shaking intensity qualitatively (I–XII)
- (B) Richter measures water temperature; Mercalli measures air humidity
- (C) Mercalli is used only for volcanic eruptions; Richter is used for tsunamis
- (D) There is no difference; both measure the number of deaths in an earthquake

Q.24 What is a 'levee' along a mature river floodplain, and how does it form naturally?

- (A) A low natural embankment of coarse silt deposited along river margins during repeated overbank floods as water velocity abruptly decelerates upon leaving the channel
- (B) A concrete wall constructed exclusively by government contractors
- (C) A deep trench dug to drain all river water into underground mines
- (D) A sand dune blown into the river channel by coastal storms

Q.25 Why is the earth's crust beneath the oceans (oceanic crust) significantly thinner and denser than the continental crust?

- (A) Oceanic crust is composed of dense mafic basalt rich in iron and magnesium (~5–10 km thick); continental crust is composed of buoyant felsic granite rich in silica and aluminum (~35–70 km thick)
- (B) Oceanic crust is made of frozen ocean salt water
- (C) Continental crust is hollow inside like a balloon
- (D) Oceanic crust was formed by falling meteorites in the year 1800

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & STEP-BY-STEP EXPLANATIONS (SET C)

Q. No.	Correct	Curricular Reference & In-Depth Conceptual Solution
Q.1	(A)	High silica content ($>60\%$) creates viscous magma that traps gas until catastrophic explosive decompression occurs, building steep pyroclastic cones; fluid basalt ($<50\%$ SiO_2) forms broad shield domes.
Q.2	(A)	Helicoidal secondary flow drives highest tangential velocities against the concave outer cut-bank (corrasion), while reduced shear velocity on convex inner banks induces alluvial point-bar deposition.
Q.3	(A)	Rift valleys are tectonic grabens dropped between divergent fault planes; gorges are fluvial erosional canyons carved through vertical hydraulic downcutting into uplifted plateaus.
Q.4	(A)	Feldspar hydrolysis: $2\text{KAlSi}_3\text{O}_8 + 2\text{H}_2\text{CO}_3 + 9\text{H}_2\text{O} \rightarrow \text{Al}_2\text{Si}_2\text{O}_5(\text{OH})_4 \text{ (Kaolinite)} + 2\text{K}^+ + 2\text{HCO}_3^- + 4\text{H}_4\text{SiO}_4$, crumbling hard granite.
Q.5	(A)	Seismic impedance contrast amplifies ground acceleration in unconsolidated alluvium, and pore-water pressure spikes induce liquefaction, turning solid ground into fluid quicksand.
Q.6	(A)	As groundwater charged with calcium bicarbonate degasses CO_2 inside caves, CaCO_3 precipitates: stalactites hang down from ceilings (*c* for ceiling); stalagmites grow up from the ground (*g* for ground).
Q.7	(A)	Plate tectonics is driven by mantle thermal convection cells and gravitational ridge-push at spreading centers, dominated by negative buoyancy 'slab-pull' of dense sinking lithospheric slabs.
Q.8	(A)	Weathering fragments bedrock *in situ*; mass wasting involves gravitational downslope transport (rockfalls, landslides, mudflows) without requiring an intermediate transporting agent like a river.
Q.9	(A)	Greater ice volume and thickness in the main valley accelerates basal glacial scouring; smaller tributary glaciers carve shallower troughs, leaving hanging valleys upon deglaciation.
Q.10	(A)	Saltation accounts for $\sim 70\text{--}80\%$ of sand transport, bouncing within 1 meter of the surface; turbulence lifts fine silt (<0.05 mm) into suspension, carrying loess dust over thousands of kilometers.
Q.11	(A)	Old Mesozoic Pacific oceanic crust (~ 170 million years old) is extremely dense and cold, descending at a steep $\sim 90^\circ$ angle at the Mariana Trench, plunging 11 km below sea level.
Q.12	(A)	Lateral tectonic compression buckles strata: upward convex folds form anticlines; downward concave depressions form synclines (though mature inverted relief can reverse this).
Q.13	(A)	Chemical weathering requires liquid water and thermal activation ($>10^\circ\text{C}$); frost wedging and thermal expansion dominate where liquid water is frozen or virtually absent.
Q.14	(A)	Alluvial fans are subaerial cone-shaped deposits formed as confined mountain streams enter flat unconfined piedmont valleys; deltas are subaqueous/subaerial coastal deposits.
Q.15	(A)	Hydraulic action, corrasion, and subaerial weathering continue to attack the isolated sea stack; when it collapses, the wave-planed remnant is termed a sea stump.
Q.16	(A)	Lichens produce chelating organic acids that chemically weather minerals, initiating the pedogenic (soil-forming) cycle on barren lithic surfaces.
Q.17	(A)	In the youthful stage, vertical corrasion dominates; the stream follows lines of least structural resistance, winding around alternating ridges of harder rock to create interlocking spurs.
Q.18	(A)	Cirques are amphitheater-like bedrock hollows with steep headwalls carved by rotational slip, plucking, and nivation beneath headward glacial ice caps.
Q.19	(A)	Tarn lakes (corrie lochs) are deep, pristine freshwater lakes occupying rock basins excavated by cirque glaciers.

Q. No.	Correct	Curricular Reference & In-Depth Conceptual Solution
Q.20	(A)	Wind lifts fine sand via deflation; the residual lag gravels form a tightly interlocked protective surface layer called desert pavement (reg or gibber plains).
Q.21	(A)	Crustal extension causes normal faulting: downthrown blocks form grabens (e.g., Rhine Rift, Narmada valley), and upthrown flanking blocks form horsts (e.g., Vosges, Black Forest).
Q.22	(A)	Pressure-release jointing (sheeting): erosion unburdens deep plutonic granite, relieving confining lithostatic pressure and causing curved concentric spalling (exfoliation domes).
Q.23	(A)	Richter/Moment Magnitude (M_w) quantifies physical radiated energy instrumentally; Modified Mercalli Intensity (MMI) scales human perception and structural damage empirically.
Q.24	(A)	When a river breaches its banks during floods, velocity drops instantly at the channel margin, dumping coarse alluvium to build natural linear ridges termed levees.
Q.25	(A)	Oceanic crust (SIMA) is dense ($\sim 3.0 \text{ g/cm}^3$) basaltic rock formed at mid-ocean ridges; continental crust (SIAL) is buoyant ($\sim 2.7 \text{ g/cm}^3$) granitic rock that floats higher on the mantle.